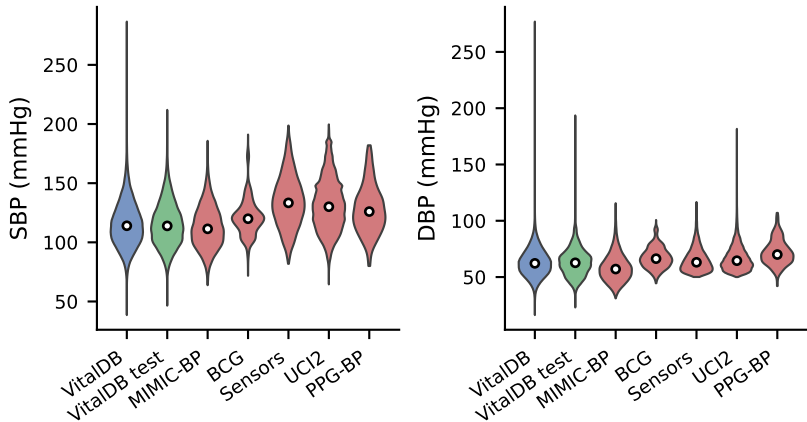


a

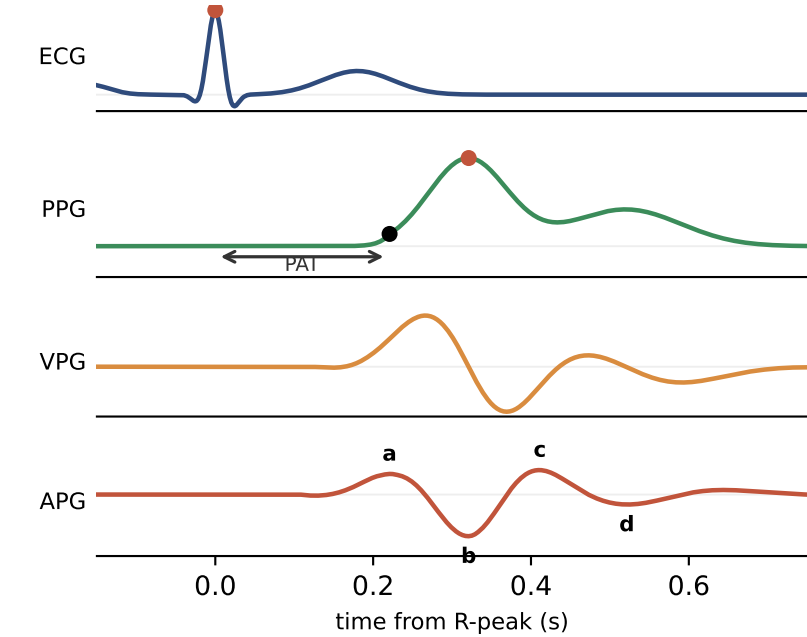


b

Datasets

Dataset	Role	Seg	Subj	SBP/DBP	Age [5-95]	Chan
VitalDB	train	465k	1,293	115/63	61 [31-79]	ECG+PPG
VitalDB test	ID	58k	144	115/63	60 [30-79]	ECG+PPG
MIMIC-BP	OOD	137k	1,524	113/58	--	ECG+PPG
BCG	OOD	3k	40	121/67	28 [19-65]	PPG
Sensors	OOD	11k	1,195	134/65	57 [33-81]	PPG
UCI2	OOD	411k	10,793	132/67	--	PPG
PPG-BP	OOD	619	218	128/72	58 [25-80]	PPG

c Feature extraction (ECG, PPG, 1st/2nd deriv.)

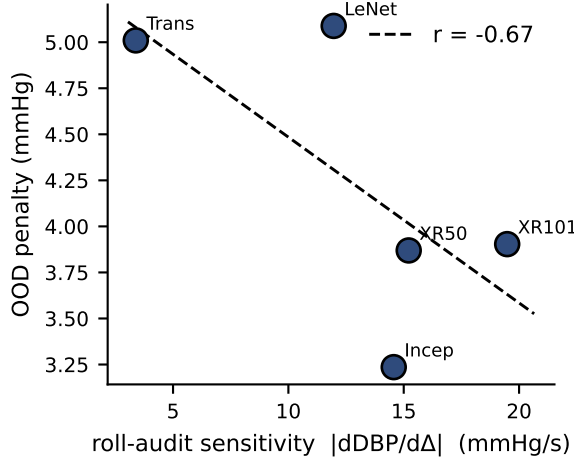


d

DBP MAE (mmHg): ID VitalDB, OOD MIMIC-BP

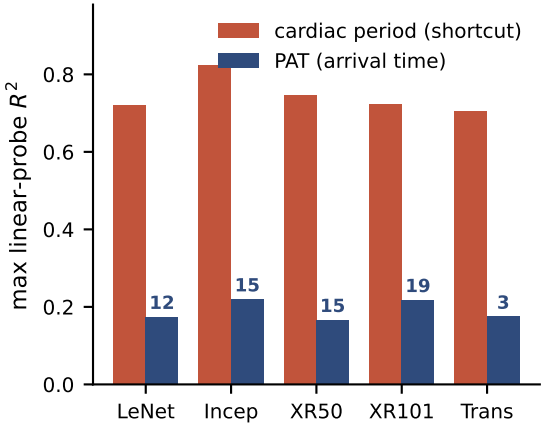
Model	size	ID	OOD
LeNet1d	659k	8.5	13.5
Inception1d	472k	8.6	11.8
XResNet50	887k	8.7	12.6
XResNet101	1.8M	8.6	12.5
Transformer	107k	8.6	13.6
LightGBM, all features	~1.8k par	8.47	15.69
LightGBM single tree, 4 feat	~190 par	8.92	12.24
LightGBM single tree, PAT only	~190 par	9.34	10.96

e Mechanism vs OOD robustness



f

Decodability is flat across models (number = |roll slope|, mmHg/s)



g

Calibration burden: demographics halve the cuff readings needed

